

SOFTWARE TESTING AND QUALITY ASSURANCE

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Test Case Design Sheet

Test Case ID	Description	Input	Expected Output	Priority
TC1	Valid construction	productId="P1", location="L1", onHand=10, threshold=5, maxCap=50	Object created successfully	High
TC2	Invalid: null productId	productId=null	IllegalArgumentException	High
TC3	Invalid: blank productId	productId=""	IllegalArgumentException	High
TC4	Invalid: null location	location=null	IllegalArgumentException	High
TC5	Invalid: blank location	location=""	IllegalArgumentException	High
TC6	Invalid: negative onHand	onHand=-1	IllegalArgumentException	High
TC7	Invalid: negative threshold	threshold=-2	IllegalArgumentException	Medium
TC8	Invalid: threshold > maxCapacity	threshold=200, maxCap=100	IllegalArgumentException	Medium
TC9	Invalid maxCapacity	maxCapacity=-5	IllegalArgumentException	High

Test Case Design Sheet

Test Case ID	Description	Input	Expected Output	Priority
TC10	Invalid: onHand > maxCapacity	onHand=60, maxCap=50	IllegalArgumentException	High
TC11	Add stock normally	addStock(10)	onHand increases by 10	High
TC12	Add stock zero	addStock(0)	onHand unchanged	Medium
TC13	Add negative stock	addStock(-5)	IllegalArgumentException	Medium
TC14	Add stock exceeding capacity	addStock(999)	IllegalStateException	High
TC15	Reserve normally	reserve(5)	reserved=5	High
TC16	Reserve more than available	reserve(999)	IllegalStateException	High
TC17	Reserve negative	reserve(-1)	IllegalArgumentException	Medium
TC18	Release normally	release(3)	reserved decreases	High

Test Case Design Sheet

Test Case ID	Description	Input	Expected Output	Priority
TC19	Release more than reserved	release(999)	IllegalStateException	Medium
TC20	Ship normally	ship(5)	onHand-=5, reserved-=5	High
TC21	Ship more than reserved	ship(50)	IllegalStateException	High
TC22	Ship negative	ship(-1)	IllegalArgumentOutOfRangeException	Medium
TC23	Remove damaged valid	removeDamaged(3)	onHand decreases	Medium
TC24	Remove damaged > onHand	removeDamaged(999)	IllegalStateException	Medium
TC25	Remove damaged negative	removeDamaged(-1)	IllegalArgumentOutOfRangeException	Low
TC26	ReorderNeeded true	available < threshold	true	High
TC27	ReorderNeeded false	available ≥ threshold	false	Medium

Test Case Design Sheet

Test Case ID	Description	Input	Expected Output	Priority
TC28	Update location	setLocation("NEW")	location updated	Low
TC29	Update location null	setLocation(null)	IllegalArgumentException	Medium
TC30	Update capacity	setMaxCapacity(200)	maxCapacity updated	Medium
TC31	Invalid negative capacity	setMaxCapacity(-1)	IllegalArgumentException	High
TC32	Capacity < onHand	setMaxCapacity(20)	IllegalStateException	Medium
TC33	Disabled test	—	Test skipped	Low
TC34	Auto-location placeholder	—	Not implemented	Low

COVERAGE

Coverage ProductStockTest				
Element ^	Class, %	Method, %	Line, %	Branch, %
all	50% (1/2)	89% (17/19)	84% (65/77)	79% (46/58)
org.example	0% (0/1)	0% (0/1)	0% (0/3)	0% (0/2)
Main	0% (0/1)	0% (0/1)	0% (0/3)	0% (0/2)
ProductStock	100% (1/1)	94% (17/18)	87% (65/74)	82% (46/56)

Cover ProductStockTest	
ProductStock Complete Test Suite	33 tests passed, 1 ignored 34 tests total, 116 ms
Constructor creates object	"C:\Program Files\Java\jdk-25\bin\java.exe" ...
Constructor rejects null or 3 ms	Starting ProductStock Complete Test Suite
Constructor rejects null or 1 ms	Test completed
Constructor rejects negative 3 ms	Test completed
Constructor rejects initial 1 ms	Test completed
Change location successfully 2 ms	Test completed